Unit 3 Numerical Integration Results Document

Ling Qingyang s2471545

Task 4.2

Cosmology with default parametres: H_0=72.0 km/s/Mpc, Ω_m=0.3, Ω_Λ=0.7, Ω_k=0.0.

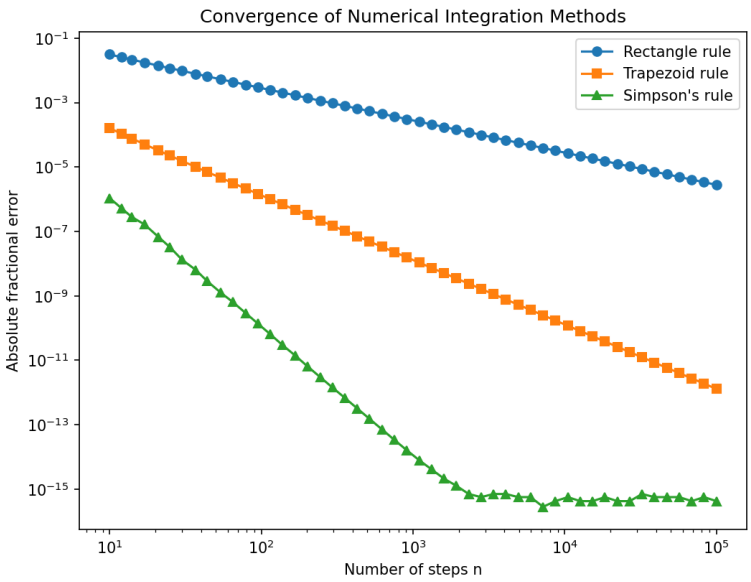
The results of three different methods of numerical integration by using a very large number of steps (n=10000) are obtained as follows:

Final Rectangle result (n=10000): 3212.1457385974804

Final Trapezoid result (n=10000): 3212.055783918915

Final Simpson result (n=10000): 3212.055783501705

The following graph illustrates the absolute fractional error in the distance as a function of the number of steps for each method. Obviously, the accuracy of numerical integrations increases as the number of steps get larger, except for Simpson' Rule which achieved maximum accuracy for computers (10^ (-15)) in just more than 1000 steps.



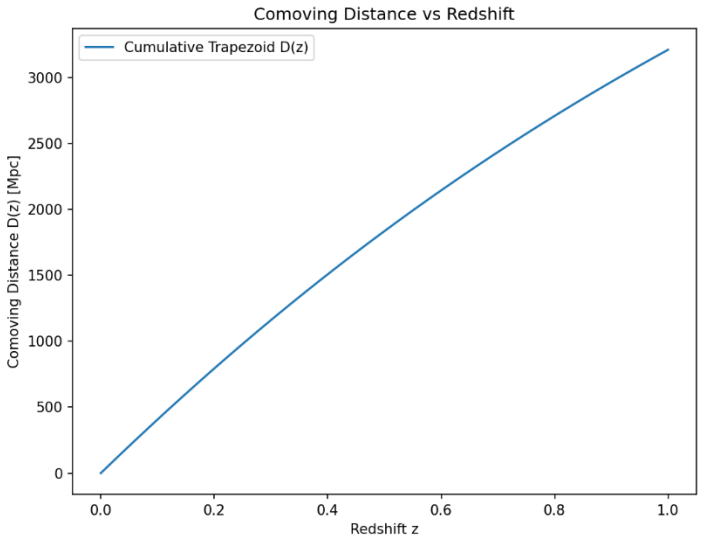
Physically, accuracies up to (10^ (-4)) or (10^ (-5)) are enough. However, in computer simulation, I insist that the more accurate, the better.

So, we should adopt the most accurate Simpson' rule with an accuracy level up to 5 significant figures: D(z=1.0) = 3212.05578 Mpc

Task 4.3

The figure below is the cumulative form of the Trapezoid rule to compute an array of distances to redshifts from 0 to 1.0.

The results obtained by using the method of interpolation that accepts an array of z values and returns an array of corresponding distances D(z), illustrated as below, perfectly matches the graph of cumulative form.



Interpolated distances:

z=0.00, D(z)=4.16 Mpc

z=0.10, D(z)=406.83 Mpc

z=0.30, D(z)=1161.22 Mpc

z=0.50, D(z)=1836.16 Mpc

z=0.70, D(z)=2436.10 Mpc

z=1.00, D(z)=3212.06 Mpc

By using the same interpolation method, distance moduli μ(z) of all the redshift values are computed and as shown as follows:

Distance moduli:

z=0.00, μ(z)=28.10

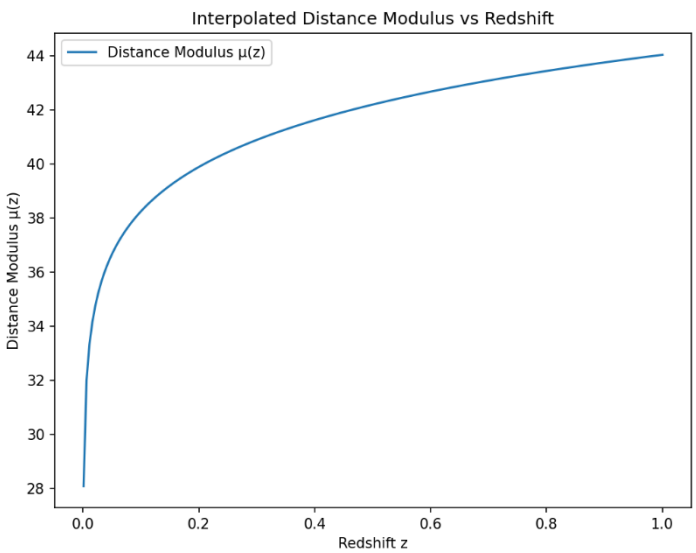
z=0.10, μ(z)=38.25

z=0.30, μ(z)=40.89

z=0.50, μ(z)=42.20

z=0.70, μ(z)=43.09

z=1.00, μ(z)=44.04

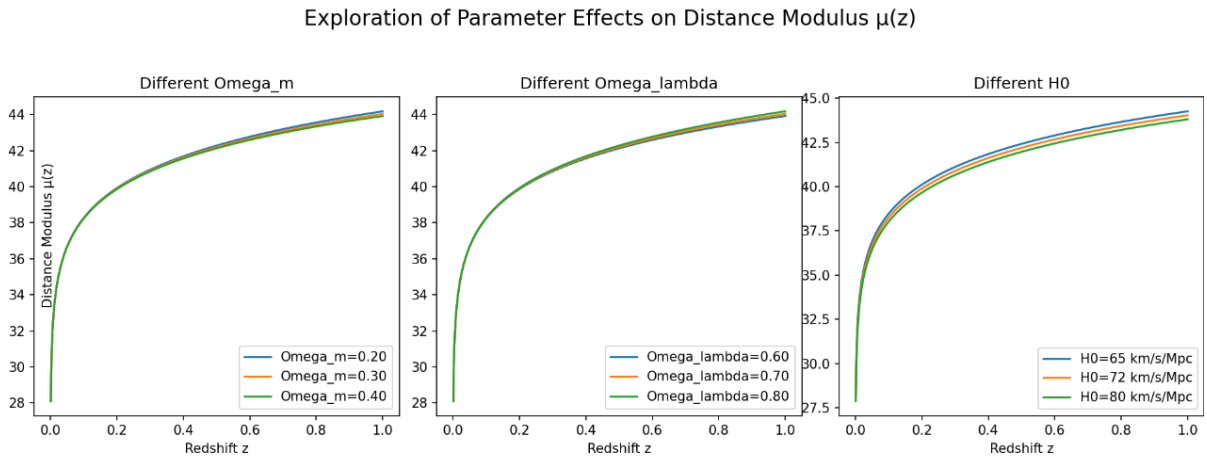


Task 4.4

By one parameter at a time while holding the others fixed, and plot $\mu(z)$to $z = 1$.

We can easily conclude from the three graphs below, that:

Varying Ω_m(e.g. 0.2/0.3/0.4): higher $\Omega_m \rightarrow$ lower $\mu(z)$.

Varying Ω_Λ(e.g. 0.6/0.7/0.8): higher $\Omega_\Lambda \rightarrow$ higher $\mu(z)$.

Varying H_0(e.g. 65/72/80 km/s/Mpc): higher $H_0 \rightarrow$ lower $\mu(z)$.